Project Management Plan

- **Feasibility study**:

  - **Technical Feasibility:** The system will be a C++ application (console or simple desktop GUI) running on a single school computer or local network. Thus, the system is Technically feasible because the required technologies are suitable for implementation of a Moderated Learning Management System.
  - **Operational Feasibility:** Every user logs in with a username and password and can only perform the actions allowed by their role. For example, the Administrator manages user accounts, departments and courses, assigns lecturers to courses and enrolls students. The principal also requires the system to be simple since *teachers are not IT experts*. So, the system will be operationally feasible provided the interface is kept simple and users are given appropriate instructions.
  - **Economic Feasibility:** The system will run on a single computer or local network, also the IT Officer expects the system to work offline since the computers are old and internet is unreliable. So, it is Economically feasible for the existing local infrastructure and does not require online payment, video conferencing, automated grading, plagiarism detection, and web or mobile interfaces.

  - **Legal Feasibility:** IT Officer states, “Passwords must be protected”, which implies passwords may be encrypted as a form of securing user passwords. All changes to marks must be recorded so that the school can who changed the marks, when and why. Hence, the system is Legally feasible provided that the application data-protection requirements are identified and the system implements appropriate security and access controls.

Schedule Feasibility:

Week	Main Activities
1	Feasibility study, work breakdown, schedule, risk register
2	Stakeholder analysis, conflict resolution, SRS, use case model
3	Architecture, detailed UML design, design patterns, database design
4	C++ development with SQLite, CI pipeline, code reviews
5	Unit, integration, system and acceptance testing
6	Packaging, installation guide, user manual, release v1.0
7	Change request, impact analysis, release v1.1, regression testing and Professional issues, final report, viva

Overall Feasibility Recommendation

The proposed Moderate Learning Management System is feasible from technical, operational, economic, legal and scheduling perspectives, subject to appropriate security, requirements management and time management. The project should therefore proceed to requirements engineering and detailed planning.

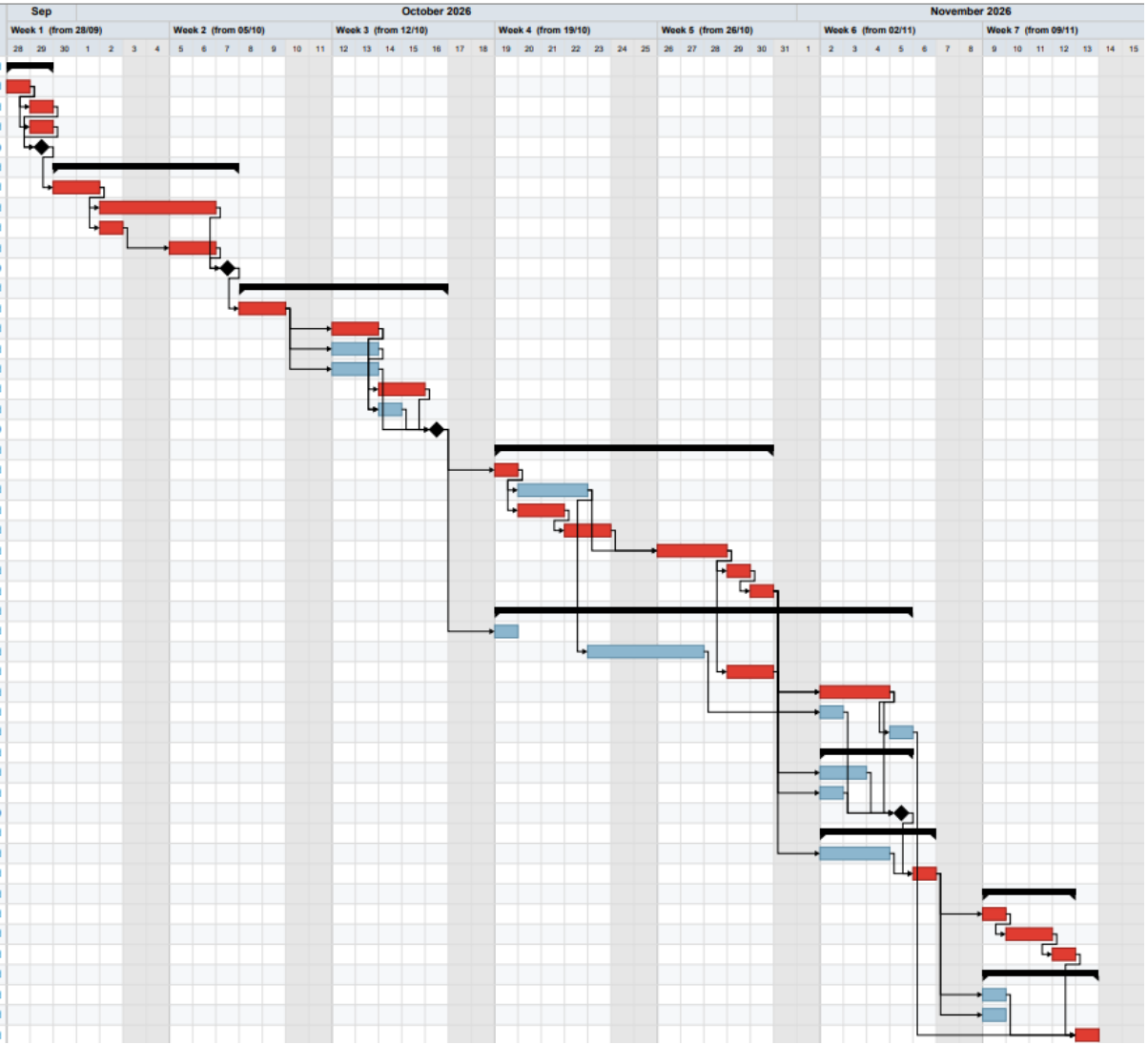
- **Work Breakdown Structure**:
  - Project Planning and management
    - Feasibility study
    - Work Breakdown Structure and Gantt chart
    - Risk register
  - Requirements engineering
    - Stakeholder analysis and conflict resolution
    - Software Requirements Specification
    - Use case diagram
    - Three fully dressed use case
  - System Design

        - Architecture
        - Class diagram and patterns
        - Database design
        - State and activity diagram
        - Sequence diagram
    - Implementation
        - Skeleton
        - CMake and CI pipeline
        - Domain model and penalty rules
        - Data-access layer and audit log
        - Authentication and RBAC
        - Role workflows and UI
        - Validation
    - Verification and Validation
        - Test plan
        - Unit tests
        - Integration tests
        - System and acceptance tests with defect log
        - Coverage report
        - Traceability matrix
    - Configuration Management and Quality Assurance
        - Static analysis, metrics and review record
        - Branching strategy and CI evidence
    - Deployment
        - Deployment diagram, manuals and release notes
        - Package and tag v1.0
    - Maintenance

        - CR1 classification and impact analysis
        - Implement CR1(V1.1)
        - Regression tests
    - Completion
        - Professional, legal and ethical section
        - Final report assembly

GANTT CHART

Name	Begin date	End date	Dur.
1 Planning and management	28/09/26	29/09/26	2d
1.1 Feasibility study	28/09/26	28/09/26	1d
1.2 Work breakdown and Gantt chart	29/09/26	29/09/26	1d
1.3 Initial risk register	29/09/26	29/09/26	1d
M1 Project plan baselined	29/09/26	29/09/26	0
2 Requirements engineering	30/09/26	07/10/26	6d
2.1 Stakeholder analysis and conflicts	30/09/26	01/10/26	2d
2.2 Software Requirements Specification	02/10/26	06/10/26	3d
2.3 Use case diagram	02/10/26	02/10/26	1d
2.4 Use case specifications (3)	05/10/26	06/10/26	2d
G1 Gate 1: requirements review	07/10/26	07/10/26	0
3 System design	08/10/26	16/10/26	7d
3.1 Architecture and components	08/10/26	09/10/26	2d
3.2 Class diagram and patterns	12/10/26	13/10/26	2d
3.3 Database design (ERD, SQL)	12/10/26	13/10/26	2d
3.4 State and activity diagrams	12/10/26	13/10/26	2d
3.5 Sequence diagrams	14/10/26	15/10/26	2d
3.6 Database-to-class mapping	14/10/26	14/10/26	1d
G2 Gate 2: design review	16/10/26	16/10/26	0
4 Implementation	19/10/26	30/10/26	10d
4.1 Skeleton, CMake and CI	19/10/26	19/10/26	1d
4.2 Domain model and penalty rules	20/10/26	22/10/26	3d
4.3 Data-access layer and audit log	20/10/26	21/10/26	2d
4.4 Authentication and RBAC	22/10/26	23/10/26	2d
4.5 Role workflows and UI	26/10/26	28/10/26	3d
4.6 Validation and code review	29/10/26	29/10/26	1d
4.7 Integration and build guide	30/10/26	30/10/26	1d
5 Verification and validation	19/10/26	05/11/26	14d
5.1 Test plan	19/10/26	19/10/26	1d
5.2 Unit tests	23/10/26	27/10/26	3d
5.3 Integration tests	29/10/26	30/10/26	2d
5.4 System and acceptance tests	02/11/26	04/11/26	3d
5.5 Coverage report	02/11/26	02/11/26	1d
5.6 Traceability matrix	05/11/26	05/11/26	1d
6 Configuration management and QA	02/11/26	05/11/26	4d
6.1 Static analysis and metrics	02/11/26	03/11/26	2d
6.2 CM and CI evidence	02/11/26	02/11/26	1d
G3 Gate 3: testing review	05/11/26	05/11/26	0
7 Deployment	02/11/26	06/11/26	5d
7.1 Deployment diagram and manuals	02/11/26	04/11/26	3d
7.2 Package and tag release v1.0	06/11/26	06/11/26	1d
8 Maintenance	09/11/26	12/11/26	4d
8.1 CR1 classification and impact	09/11/26	09/11/26	1d
8.2 Implement CR1 (release v1.1)	10/11/26	11/11/26	2d
8.3 Regression tests, tag v1.1	12/11/26	12/11/26	1d
9 Completion	09/11/26	13/11/26	5d
9.1 Professional and ethical issues	09/11/26	09/11/26	1d
9.3 Viva preparation	09/11/26	09/11/26	1d
9.2 Final report assembly	13/11/26	13/11/26	1d



RISK MANAGEMENT

ID	Risk	Category	Likelihood	Impact	Exposure	Mitigation	Owner
R1	Requirements may be unclear or conflicting between stakeholders	Project	Medium	High	High	Analyze stakeholder statements early, document conflicts and obtain agreement on resolved requirements	Project Manager
R2	C++ implementation may take longer than planned	Technical	Medium	Very high	Very high	Develop incrementally, follow the approved design and implement core functionality first	Developer
R3	Database errors or incorrect database design may cause data loss or inconsistent records	Technical	Low	Very high	Medium	Design the database carefully, use constraints, transactions and regular database testing	Developer
R4	Security weaknesses could expose passwords or assessment records	Technical	Low	Very high	Medium	Use salted password hashes, role-based access control and parameterized SQL queries	Developer
R5	Project tasks may fall behind the seven-week schedule	Project	Medium	Very high	Very high	Use the Gantt chart, monitor milestones and	Project Manager

						prioritize critical-path activities	
R6	Testing may reveal major defects late in the project	Technical	Medium	High	High	Start automated testing during implementation and test each major feature incrementally	Tester
R7	Git/CI configuration problems may prevent reliable builds or testing	Technical	Low	Medium	Low	Set up Git and CI early and test the pipeline with small commits	Developer
R8	Users may find the system difficult to use	People/Operational	Medium	Medium	Medium	Keep the interface simple and provide user manuals for each role	Project Manager